

# Load Interconnection Request System Impact Study Report

## For Queue Project Q14-0302

George M Sullivan Generation Plant 2 100 kV  
June 27, 2015

### **1. Introduction**

This System Impact Study report documents the results of the analyses performed for interconnection request Q14-0302. The Interconnection Customer proposes a 229.5 MW large load (data center) facility interconnecting at the George M Sullivan Generation Plant 2 100 kV substation in the Illinois area. The request was received on January 11, 2014. The study evaluated system intact and single contingency conditions using the regional planning case, including 188 higher-queued active requests and their assigned network upgrades.

### **2. Summary of Results**

The project was found to contribute (DFAX of 5% or greater) to 11 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$33,604,000.

### **3. Network Impacts**

#### **3.1 Load Deliverability Analysis**

1. Facility: GEORGE M SULLIVAN GENERATION PLANT 2 – CRUM CANYN 100 kV Line  
Contingency: JONESBORO SPA SUBSTATION – PINE BLUFF S 100 kV  
Loading: 239.0% (Rating: 123 MVA)  
DFAX: 100.0%  
Pre-project loading: 52.6%
2. Facility: GEORGE M SULLIVAN GENERATION PLANT 2 – PINE BLUFF S 100 kV L  
Contingency: GEORGE M SULLIVAN GENERATION PLANT 2 – CRUM CANYON 100 kV  
Loading: 230.8% (Rating: 109 MVA)  
DFAX: 100.0%  
Pre-project loading: 19.6%
3. Facility: JONESBORO SPA – PINE BLUFF SOUTH 100 kV  
Contingency: GEORGE M SULLIVAN GENERATION PLANT 2 – CRUM CANYON 100 kV  
Loading: 222.2% (Rating: 132 MVA)  
DFAX: 100.0%  
Pre-project loading: 48.9%
4. Facility: JONESBORO SPA SUBSTATION – PADUA 100 kV line  
Contingency: GEORGE M SULLIVAN GENERATION PLANT 2 – CRUM CANYON 100 kV L  
Loading: 176.5% (Rating: 133 MVA)  
DFAX: 76.3%  
Pre-project loading: 44.5%
5. Facility: WASHINGTON PARK – SPRING CITY 230 kV line  
Contingency: EVERGREEN PARK – FISK STREET 230 kV  
Loading: 138.3% (Rating: 551 MVA)  
DFAX: 6.0%  
Pre-project loading: 135.8%

6. Facility: PADUA SUBSTATION – RIO BRAVO ROCKLIN 100 kV  
Contingency: RIO BRAVO ROCKLIN SUBSTATION – BAKER STREET SUBSTATION 100 kV line  
Loading: 121.7% (Rating: 140 MVA)  
DFAX: 32.2%  
Pre-project loading: 68.9%
7. Facility: BAKER STREET – RIO BRAVO ROCKLIN 100 kV Line  
Contingency: PADUA – RIO BRAVO ROCKLIN 100 kV Line  
Loading: 114.0% (Rating: 141 MVA)  
DFAX: 29.2%  
Pre-project loading: 66.4%
8. Facility: TWIN COUNTY RANCH – SWAN VALLEY 230 kV L  
Contingency: SAN VALLEY – TREGO 500 kV L  
Loading: 110.4% (Rating: 773 MVA)  
DFAX: 7.3%  
Pre-project loading: 108.2%
9. Facility: CRUM CANYON – NORTH CHARLOTTE 100 kV line  
Contingency: SANTA TERESA – PINHOOK 100 kV  
Loading: 109.8% (Rating: 139 MVA)  
DFAX: 45.4%  
Pre-project loading: 34.8%
10. Facility: CRUM CANYON – PADUA 100 kV  
Contingency: JONESBORO SPA – PINE BLUFF S SUBSTATION 100 kV Line  
Loading: 109.0% (Rating: 123 MVA)  
DFAX: 39.1%  
Pre-project loading: 35.9%
11. Facility: PADUA – BAKER STREET 100 kV  
Contingency: PADUA SUBSTATION – RIO BRAVO ROCKLIN 100 kV Line  
Loading: 103.9% (Rating: 133 MVA)  
DFAX: 29.2%  
Pre-project loading: 53.5%

### **3.2 Multiple Facility Contingency Analysis**

No additional overloads were identified for multiple facility contingencies.

### **3.3 Short Circuit and Stability**

Short circuit and stability analyses identified no additional network upgrades.

## **4. Required Network Upgrades and Cost Estimates**

| Facility                                                      | Required Upgrade                            | Cost Estimate | Allocation to Project |
|---------------------------------------------------------------|---------------------------------------------|---------------|-----------------------|
| GEORGE M SULLIVAN GENERATION PLANT 2 – CRUM CANYN 100 kV Line | Rebuild line with higher-capacity conductor | \$3,354,000   | 100% (\$3,354,000)    |
| PINE BLUFF S – GEORGE M SULLIVAN GENERATION PLANT 2 100 kV    | Rebuild line with higher-capacity conductor | \$12,118,000  | 100% (\$12,118,000)   |
| JONESBORO SPA – PINE BLUFF SOUTH 100 kV                       | Rebuild line with higher-capacity conductor | \$4,654,000   | 100% (\$4,654,000)    |
| JONESBORO SPA SUBSTATION – PADUA 100 kV                       | Rebuild line with higher-capacity conductor | \$3,451,000   | 100% (\$3,451,000)    |
| WASHINGTON PARK – SPRING CITY 230 kV                          | Rebuild line with higher-capacity conductor | \$7,430,000   | 2% (\$179,000)        |

| Facility                                       | Required Upgrade                               | Cost Estimate | Allocation to Project |
|------------------------------------------------|------------------------------------------------|---------------|-----------------------|
| PADUA SUBSTATION – RIO BRAVO<br>ROCKLIN 100 kV | Rebuild line with higher-capacity<br>conductor | \$2,207,000   | 70% (\$1,545,000)     |
| RIO BRAO ROCKLIN – BAKER ST 100<br>kV          | Rebuild line with higher-capacity<br>conductor | \$2,502,000   | 70% (\$1,752,000)     |
| TWIN COUNTY RANCH – SWAN<br>VALLEY 230 kV L    | Rebuild line with higher-capacity<br>conductor | \$47,591,000  | 3% (\$1,505,000)      |
| CRUM CANYON – N CHARLOTTE 100<br>kV            | Reconductor line                               | \$2,922,000   | 88% (\$2,569,000)     |
| CRUM CANYON – PADUA 100 kV                     | Reconductor line                               | \$2,336,000   | 68% (\$1,588,000)     |
| PADUA – BAKER STREET 100 kV                    | Reconductor line                               | \$1,271,000   | 70% (\$889,000)       |
|                                                | Total allocated to project                     |               | \$33,604,000          |

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.